Ming-Ching Chang

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Albany, NY 12222, USA

Education

9/2001-9/2008 **Ph.D.**, Laboratory for Engineering Man/Machine Systems (LEMS), Division of Engineering, Brown University, USA. GPA: 3.93/4.0.
Thesis: *The Medial Scaffold for 3D Shape Modeling and Recognition.*

9/1996-6/1998 **Master of Science**, Dept. of Computer Science and Information Engineering (CSIE), National Taiwan University (NTU), Taiwan. GPA: 90.17/100 (3.89/4.0).
Thesis: *Fast Search Algorithms for IC Printed Mark Quality Inspection.*

9/1992-6/1996 **Bachelor of Science in Engineering**, Dept. of Civil Engineering, National Taiwan University.

Affiliation / Experience

4/2026-present *Chief Executive Officer,* **Elyrix Inc.** (Start-up), USA.

8/2024-1/2025 *Visiting Scholar*, Dept. of Computer Science and Information Engineering, **National Taiwan University**, Taiwan.

5/2021-present *Senior Principal Research Advisor*, AI Research Center, **Inventec Corp.**, Taiwan.

7/2021-present *Visiting Associate Professor*, Center for GIS, RCHSS, **Academia Sinica**, Taiwan.

9/2022-present *Associate Professor* (Tenured), Computer Science, College of Nanotechnology, Science, and Engineering (CNSE), **University at Albany, State University of New York (SUNY)**.

2/2021-7/2021 *Visiting Assistant Professor***,** Dept. of Computer Science, **National Central University**, Taiwan.

7/2018-9/2022 *Assistant Professor*, Computer Science, University at Albany, State University of New York (SUNY).

9/2016-7/2018 *Assistant Professor*, Dept. of Electrical and Computer Engineering (ECE), University at Albany, State University of New York (SUNY).

11/2013-9/2016 *Lead Computer Scientist*, Computer Vision Lab., **GE Global Research Center**

11/2008-11/2013 *Computer Scientist*, Computer Vision Lab., GE Global Research Center

9/2012-9/2016 *Adjunct Professor*, Dept. of Computer Science, University at Albany, State University of New York (SUNY).

2001-9/2008 *Research Assistant*, Laboratory for Engineering Man/Machine Systems (LEMS), Division of Engineering, **Brown University**, USA.

8/1999-6/2001 *Teaching Assistant*, Dept. of Computer Center of College of Management, **National Taiwan University**.

7/1998-7/1999 *Mandatory Military Service*, Air force logistics, Taiwan.

9/1996-6/1998 *Research Assistant*, Mechanical Industry Research Labs. (MIRL), **Industrial Tech. Research Inst. (ITRI)**, Taiwan.

6/1997-6/1998 *R&D Engineer*, UTechZone Co., Ltd.

Research Programs/Grants (All amounts are list in USD$)

1. **PI,** *UAlbany Strategic Partnership for Industrial Resurgence (SPIR) fund* 2026-2027
2. **PI,** *PathScience research fund* 2026-2027
3. **Co-PI**, Embedded Edge Computing Framework for Real-Time Road Object Tracking for Wide Coverage using Deep Learning (PI: Munkhjargal Gochoo), UAE University 2024-2027
4. **PI**, *Secure Artificial Intelligence with Fully Homomorphic Encryption for Deep Neural Networks* ($10K), University at Albany Faculty Research Awards Program. 2023-2024
5. **Co-I**, *Artificial Intelligence Institute for Secure and Trustworthy Cyber-Systems* ($20K), **UAlbany** Seed Fund. 2022
6. **Co-I**, *Center for Integrated Sensors, Screening, and Security (CIS3)* ($20K), **UAlbany** Seed Fund. 2022
7. **Co-PI**, *AI Cup Challenge on Drone Object Detection and Counting* (~$130K), National Yang Ming Chiao Tung University, **Ministry of Education**, Taiwan. 2021-2023
8. **Co-PI**, *AI Cup Challenge on Chinese Scene Text Detection and Recognition* (~$130K), National Central University, **Ministry of Education**, Taiwan. 2020-2022
9. **PI**, *Semantic Information Defender* (UAlbany sub-contract $339K), Kitware Inc. **Defense Advanced Research Projects Agency (DARPA)** SemaFor Program (PI: Arslan Basharat) (total $13M). 2020-2024
10. **PI**, *Yard Locomotive Detection System Development* ($50K), **GE Global Research**. 2020
11. **PI**, *Yard Locomotive Detection System Development* ($200K), **GE Global Research**. 2019
12. **PI**, *Video Analytical Open Framework for Smart Transportation* ($10K) University at Albany Faculty Research Awards Program. 2019-2022
13. **PI**, *Yard Locomotive Vision System Development* ($10K), **GE Global Research**. 2018
14. **Co-PI**, *A General, Intelligent, Bio-inspired Computing Framework for Sensor Networks* ($25K, PI: Tolga Soyata), **SUNY Center-Scale Proposal Program**. 2018
15. **PI,** *Data and Algorithms for Technical Assessment of DIVA* (SUNY sub-contract $75K), **Kitware Inc.** IARPA DIVA Program (PI: Anthony Hoogs). 2018-2019
16. **PI**, *Develop Deep Learning Capability and Explanability* ($25K), **GE Global Research** 2017
17. University at Albany – SUNY Startup Funding. 2016-2021
18. **Co-PI**, *Multi-modality Portable Systems for Pressure Ulcer Prevention and Care* (total $2.75M, 2015-2016 portion $500K, PI: Peter Tu), **GE Global Research**, **Department of Veterans Affairs (VA)**, Innovation Initiative (VAi2). 2015-2016
19. **Co-PI**, *Image Analysis Platform Development for EOE Fiber Ctg Imaging* ($100K, PI: Alberto Santamaria-Pang), **GE Global Research** and **GE Aviation**. 2015-2016
20. **Co-Investigator**, *Social Interaction Analysis at a Distance* ($800K, PI: Peter Tu) GE Global Research, **Defense Advanced Research Projects Agency (DARPA)**, affiliated with the Strategic Social Interaction Modules (SSIM) program. 2013-2016
21. **Co-PI**, *Practitioner Centric Video Analytics* ($475K, PI: Peter Tu), GE Global Research, **National Institute of Justice (NIJ)** 2013-IJ-CX-K010. 2013-2015
22. **Co-PI**, *Advanced Behavior Recognition in Crowded Environments – Continuation* ($250K, PI: Peter Tu), GE Global Research, **National Institute of Justice (NIJ)** 2011-IJ-CX-K004. 2011-2013
23. **Co-PI**, *Advanced Behavior Recognition in Crowded Environments* ($500K, PI: Xiaoming Liu) GE Global Research, **National Institute of Justice (NIJ)** 2009-SQ-B9-K013. 2010-2011
24. GE Technical Career Path (TCP) Grant ($7,000) 2013
    GE Global Research Growth Council Grant ($10,000) 2010
    Developed a video content analysis prototype demonstration system.

Awards and Honors

1. **SUNY Empire AI Google GPU Service Award, 2026.**
2. **UAlbany Google Cloud GPU Award, University at Albany, 2026.**
3. **IEEE Signal Processing Society Top 25 Downloaded Articles Recognition (2024-2025), for**

"Underwater Image Enhancement and Attenuation Restoration Based on Depth and Backscatter Estimation," IEEE Transactions on Computational Imaging.

4. **Best Paper Runner-Up Award**, IEEE International Conference on Advanced Visual and Signal-based Systems (AVSS) 2025
5. **Best Paper Award**, ECCV Beyond Euclidean Workshop (BEW): Hyperbolic and Hyperspherical Learning for Computer Vision, European Conference on Computer Vision (ECCV) 2024
6. **2nd Place Winner**, IEEE Int'l Conference on Multimedia and Expo (ICME) Grand Challenge. Multi-Modal Video Reasoning and Analyzing Competition (MMVRAC) Animal Action Recognition 2024
7. **Best Student Paper Award**, IEEE Multimedia Information Processing and Retrieval (MIPR) 2023
8. National Science and Tech. Council (NSTC) grant for visiting science and technology personnel. 2023
9. Ministry of Science and Tech. (MOST) grant for short-term international visiting scholars, Yuan Ze University, Taiwan 2022
10. Ministry of Science and Tech. (MOST) grant for short-term international visiting scholars, National Central University, Taiwan 2021
11. **Best Paper Award**, National Conference on Web Intelligence and Applications (NCWIA), Taiwan 2021
12. Best Paper Finalists, IPPR Computer Vision, Graphics, and Image Processing (CVGIP), Taiwan 2020
13. Winner Teams Top-3 Finalist, AI City Challenge CVPR 2019 Workshop 2019
14. **Honorary Mention Award**, IEEE Smart World NVIDIA AI City Challenge 2017
15. NVIDIA GPU Grant Award 2017
16. **GE Belief – Stay Lean and Go Fast Management Award** 2015
17. GE Above and Beyond – Gold Award for Stay Lean to Go Fast ($2500) 2015
18. GE Above and Beyond – Bronze Award for Stay Lean to Go Fast ($100) 2015
19. GE Above and Beyond – Bronze Award for Empower and Inspire Each Other ($100) 2015
20. GE Above and Beyond – Bronze Award for External Focus Award ($250) 2015
21. GE Above and Beyond – Bronze Award for Expertise ($250) 2013
22. GE Above and Beyond – Bronze Award for Imagination & Courage ($250) 2012
23. GE Above and Beyond – Patent Award on Expertise ($275) 2012
24. GE Above and Beyond – Patent Award on Expertise ($275) 2012
25. **Best Student Paper Award**, IEEE Winter Conf. on Applications of Computer Vision (WACV) 2012
26. **Best Paper Award – Runner-Up**, IEEE Advanced Video and Signal-Based Surveillance (AVSS) 2011
27. GE Above and Beyond – Patent Award on Expertise ($275) 2011
28. GE Above and Beyond – Patent Award on Expertise ($275) 2011
29. GE Above and Beyond – Patent Award on Expertise ($275) 2011
30. GE Above and Beyond – Patent Award on Expertise ($275) 2011
31. GE Above and Beyond – Silver Award for Expertise ($500) 2011
32. GE Above and Beyond – Level 3 Award for Innovation ($100) 2010

Student Advising

Advised Ph.D. Graduated:

- Yuwei Chen, Ph.D., CS, SUNY Albany (2020-2025), Thesis: *Towards Human Explainable Digital Forensics: Generating Human Interpretable Evidence for Semantic Understanding in Manipulated Images and Text*.
- Hui Guo, Ph.D., CS (2020-2022), Thesis: *Exposing GAN-Generated Faces Using Deep Neural Network*.
- Ehab AlBadawy, Ph.D., ECE (2020-2022), Thesis: *AI-Synthesized Speech Generation and Detection*, first job after graduation: Meta Inc., Menlo Park, CA.
- Yi Wei, Ph.D., CS (2016-2021), Thesis: *Understanding Complex Human Activities in Videos*, first job after graduation: Samsung Research, Mountain View, CA.
- Shengkun Li, Ph.D., CS (2016-2020), Thesis: *Learning Graphs for Object Tracking and Counting*.

Current Ph.D. Students:

- Ting-Yu Tsai, Ph.D. student, SUNY Albany (2023-present)
- An Yu, Ph.D. student, SUNY Albany (2022-present)
- Zhenfei Zhang, Ph.D. student, SUNY Albany (2021-present)
- Rui Wang, Ph.D. candidate, CS, SUNY Albany (2019-present)
- Abhineet Pandey, MS/Ph.D. student, CS, SUNY Albany (2018-present)

Current MS Students (Selected):

- Jacob DeRosa (2025-present)
- Srujan Vepuri (2022-present)
- Ragini Rani (2022-present)
- Nidhi Vadnere (2022-present)

Co-Advised PostDoc:

- Dawei Du, CS, Albany (2017-2019), first job after completion: Kitware Inc., Clifton Park, NY. Co-advised with Siwei Lyu.
- Longyin Wen, CS, Albany (2016-2017): first job after completion: JD AI Research, Mountain View, CA. Co-advised with Siwei Lyu.

Co-Advised Ph.D. Graduated:

- Shu Hu, CS, SUNY Albany (2018-2019) and SUNY Buffalo (2019-2022), Co-advised with Siwei Lyu, first job after graduation: post doctorial researcher at CMU.
- Lipeng Ke, CS, SUNY Albany (2015-2019) and Ph.D. SUNY Buffalo (2019-2022), Co-advised with Siwei Lyu, first job after graduation: Amazon, Sunnyville, CA.
- Yuezun Li, Ph.D., CS, SUNY Albany (2015-2020), Co-advised with Siwei Lyu, Thesis: *Detecting and Protecting Against AI-Synthesized Faces*, first job after graduation: Assistant Professor, Ocean University of China.
- Wenbo Li, Ph.D., CS, SUNY Albany (2015-2019), Co-advised with Siwei Lyu, Thesis: *Pose Based Human Activity Recognition: Past, Present and Future*, first job after graduation: Samsung Research, Mountain View, CA.
- Yueming Yang, Ph.D., CS, SUNY Albany (2013-2016), Co-advised with Siwei Lyu, Thesis: *Efficient Large-Scale Photometric Reconstruction Using Divide-Conquer and Fusion*, first job after graduation: Assistant Professor, School of Mount Holyoke, MA.
- Xin Wang, Ph.D., CS, SUNY Albany (2014-2015), Co-advised with Siwei Lyu, Thesis: *Topic Analysis and Application Using Nonnegative Matrix Factorization (NMF),* first job after graduation: CuraCloud Corporation, Seattle, WA.

Ph.D. Committee Graduated:

- Ahmed Khamis, Ph.D., ECE, SUNY Albany (2025), Thesis: *Graph-based Machine Learning Framework for Power Electronic Converter Circuit Analysis*.
- Omkar Kulkarni, Ph.D., CS, SUNY Albany (2025), Thesis: *Bias-aware Gaze Uniformity Assessment in Group Images*.
- Siqian Zhao, Ph.D., CS, SUNY Albany (2024), Thesis: *Multi-Activity Student Knowledge and Behavior Modeling via Transfer Learning*.
- Miley Yao, Ph.D., CS, SUNY Albany (2021-2022), Thesis: *High-Capacity and Interpretable Temporal Point Process Models for User Activity Sequence Modeling*, first job after graduation: Meta Inc., New York, NY.
- Hadi Ghahremannezhad, Ph.D., CS, NIJT (2021), Thesis: *Advanced Traffic Video Analytics for Robust Traffic Accident Detection*.
- Sourav Dutta, CS, SUNY Albany (2022), Thesis: *Motion Planning Under Uncertainties*.

- Mehrded Mizaei, Ph.D., CS, SUNY Albany (2020), Thesis: *Discriminative Factorization Models for Student Behavioral Pattern Detection and Classification*.
- Amanda Danko, Ph.D., CS, SUNY Albany (2015), Thesis: *Bottom-Up Saliency Models Based on Natural Image Statistics*, first job after graduation: USAA Research.

Current Co-Advising Ph.D. Students or Committee:

MS Student Graduated:

- Nenghui Song, MS, CS, SUNY Albany (2017-2018).
- Jianting Wen, M.S., CS, SUNY Albany (2011-2012).

GE Research Center Intern Mentoring

- Yelin Kim, GE Research Intern (2014), Ph.D. candidate in Computer Science, University of Michigan, Ann Arbor.
- Yimeng Zhang, GE Research Intern (2011), Ph.D. in Cornell University, graduated 2012. First job: Google Mountain View.
- Karthik Sankaranarayanan, GE Research Intern (2010), Ph.D. in Ohio State University, graduated 2011. First job: IBM Research, Bangalore, India.

UAlbany Independent Study, Interns, and Mentoring

- Joshua Cho, UAlbany Honor College Student (2022-present).
- Patrick Dai, high school intern (2021).
- Ellisa Khoja, MS, CS, SUNY Albany, Independent Study for Intern at Edwards Lifesciences LLC (2018).
- Debosmita Chakraborty, M.S., CS, SUNY Albany, Independent Study for Intern at Data Core Systems, Inc (2018).
- Rutuja Shah, MS, CS, SUNY Albany, Independent Study for Intern at Center for Technology in Government, UAlbany (2018).
- Sumit Chhabra, MS, CS, SUNY Albany, Independent Study for Intern at NIIT Technologies, Inc (2018).
- Nikolas N. Aguilar, visiting student, CS, SUNY Albany (4/2015-8/2015).

Referred Journal Publications

† Students or post-docs advised or co-advised by Ming-Ching Chang are denoted by superscript (†).
* indicates the corresponding author.

1. C.-I. Huang, W.-Y. Chen, J.-W. Hsieh, P.-Y. Chen, M.-C. Chang, M. Gochoo, C.-M. Tsai, H.-W. Huang, V. S. Tseng, J.-N. Hwang, "FE-DETRAC: An Infrastructure Fisheye Video Benchmark with Distortion-Aware Hybrid Data Association for Multi-Object Tracking," IEEE Internet of Things (IoT) Journal, Apr. 2026.
2. A. Yu†, W. Lu, J. Li, Z. Zhang†, Y. Shen, F. Ye, M.-C. Chang, "SMART: Shot-Aware Multimodal Video Moment Retrieval with Audio-Enhanced MLLM", *IEEE Trans. on Multimedia*, 2026.
3. X. Wang, T.-Y. Tsai†, L. Lin, H. Guo, S. Hu, M.-C. Chang, P. Atrey, S. Lyu, "Spotting the Fakes: A Deep Dive into GAN-Generated Face Detection", *ACM Trans. On Multimedia Computing, Communications and Applications (TOMM)*, Jul. 2025.
4. Y.-T. Ku, F.-H. Liu, C.-F. Hsu, M.-C. Chang, S.-H. Hung, I-P. Tu, W.-C. Chen, "Optimizing Encrypted Neural Networks: Model Design, Quantization and Fine-Tuning Using FHEW/TFHE", *Proceedings on Privacy Enhancing Technologies (PoPETs)*, Oct. 2025.
5. C.-H. Tung, L. Ding, M.-C. Chang, G.-R. Huang, L. Porcar, Y. Wang, J.-M. Carrillo, B. Sumpter, Y. Shinohara, C. Do, W.-R. Chen, "Scattering-Based Structural Inversion of Soft Materials via Kolmogorov-Arnold Networks", *Journal of Chemical Physics*, Feb. 2025.
6. Y.-K. Hsieh, J.-W. Hsieh*, X. Li, Y.-M. Chang, Y.-C. Tseng, M.-C. Chang, "Scale-Aware Crowd Counting Network with Annotation Error Modeling", *IEEE Transactions on Image Processing (TIP)*, 2025.

7. A. Yu†, M. Singh, A. Pandey†, E. Dybas, A. Agarwal, Y. Kao, G. Zhao, T.-J. Kao, X. Li, D. Shin, M.-C. Chang*, "Integrating Manual Preprocessing with Automated Feature Extraction for Improved Rodent Seizure Classification", *Epilepsy and Behavior*, Apr. 2025.
8. J.-H. Wang, D.-Y. Wang, H.-M. Liao, M.-C. Chang, R. T.-H. Tsai*, "Unsupervised Domain Adaptation for Cross-Style, Cross-Year Land Use Understanding from Historical Maps", *International Journal of Geographical Information Science (IJGIS)*, Aug. 2025.
9. Y.-Z. Hsieh* and M.-C. Chang, "Underwater Image Enhancement and Attenuation Restoration based on Depth and Backscatter Estimation", *IEEE Transactions on Computational Imaging (TCI)*, Feb. 2025 (top 25 downloaded articles from 2024 - 2025 for IEEE TCI on IEEE Xplore).
10. T. Wang, Y. Zhao*, M.-C. Chang, J. Liu, "Open-set Occluded Person Identification with mmWave Radar", *IEEE Transactions on Mobile Computing (TMC)*, Jun. 2025.
11. Y. Xiao, F.-H. Liu, Y.-T. Ku, M.-C. Ho, C.-F. Hsu, M.-C. Chang, S.-H. Hung, W.-C. Chen, "GPU Acceleration for FHE Bootstrapping", *Transactions of Cryptographic Hardware and Embedded Systems (TCHES)*, 2024.
12. C.-H. Tung, M.-Z. Chen, H.-L. Chen, G.-R. Huang, L. Porcar, M.-C. Chang, J.-M. Carrillo, Y. Wang, B. Sumpter, C. Do and W.-R. Chen*, "Inferring Structure Factors of Charge Stabilized Colloids from Scattering Using Deep Learning", *Journal of Applied Crystallography*, vol. 57, pp. 1047-1058, 2024.
13. C.-H. Tung, Y.-J. Hsiao, H.-L. Chen, G.-R. Huang, L. Porcar, M.-C. Chang, J.-M. Carrillo, Y. Wang, B. Sumpter, Y. Shinohara, J. Taylor, C. Do, W.-R. Chen*, "Unveiling Mesoscopic Structures in Distorted Lamellar Phases Through Deep Learning-based Small Angle Neutron Scattering Analysis", *Journal of Colloid and Interface Science*, 659, 739-750, April 2024.
14. J.-A. Chen, J.-C. Hou, R. T.-H. Tsai*, H.-M. Liao, S.-P. Chen, and M.-C. Chang*, "Image Classification for Historical Documents: A Study on Chinese Local Gazetteers", *Digital Scholarship in the Humanities*, 2023.
15. I. Farady, C.-H. Lin*, M.-C. Chang, "PreAugNet: Improve Data Augmentation for Industrial Defect Classification with Small-Scale Training Data", *Journal of Intelligent Manufacturing*, pp. 179, Mar. 2023.
16. C.-H. Tung, S.-Y. Chang, M.-C. Chang, J.-M. Carrillo, B. G Sumpter, C. Do, W.-R. Chen*, "Inferring Colloidal Interaction from Scattering by Machine Learning", *Elsevier Carbon Trends*, vol. 10, 100252, Mar. 2023.
17. J.-W. Hsieh*, C.-H. Chou, M.-C. Chang, P.-Y. Chen, S. Santra, C.-S. Huang, "Mean-Shift Based Differentiable Architecture Search", *IEEE Trans. on Artificial Intelligence (TAI)*, 2022.
18. C.-S. Wong, H.-M. Liao, R. T.-H. Tsai*, M.-C. Chang*, "Semi-supervised Learning for Topographic Map Analysis Over Time: a Study of Bridge Segmentation", *Nature Scientific Report*, 12 (18997), 2022.
19. H. Guo†, S. Hu, X. Wang, M.-C. Chang*, S. Lyu, "Robust Attentive Deep Neural Network for Detecting GAN-Generated Faces", *IEEE Access*, vol. 10, pp. 32574-32583, 2022.
20. L. Ke†, M.-C. Chang*, H. Qi, S. Lyu, "DetPoseNet: Improve Multi-Person Pose Estimation via Coarse-Pose Filtering", *IEEE Transactions on Image Processing (TIP)*, vol. 31, pp. 2782-2795, 2022.
21. M.-C. Chang, C.-H. Tung, S.-Y. Chang, J. M. Carrillo, Y. Wang, B. Sumpter, G.-R. Huang, C. Do, W.-R. Chen*, "A Machine Learning Inversion Approach for Determining Effective Interaction Potential from Scattering", *Communications Physics - Nature*, 2022.
22. W. Li†, Y. Wei†, S. Lyu, M.-C. Chang*, "Simultaneous Multi-Person Tracking and Activity Recognition based on Cohesive Cluster Search", *Computer Vision and Image Understanding (CVIU)*, Elsevier, vol. 214, 2022.
23. M.-C. Su, C.-T. Cheng, M.-C. Chang, Y.-Z. Hsieh*, "A Video Analytic Student Concentration Monitoring System", *IEEE Transactions on Consumer Electronics*, Vol. 67 (4), pp. 294-304, 2021.
24. P.-Y. Chen, M.-C. Chang, J.-W. Hsieh*, Y.-S. Chen, "Parallel Residual Bi-Fusion Feature Pyramid Network for Accurate Single-Shot Object Detection", *IEEE Transactions on Image Processing (TIP)*, vol. 30, pp. 9099-9111, 2021.
25. Y. Li†, M.-C. Chang, P. Sun, H. Qi, J. Dong, S. Lyu*, "TransRPN: Towards the Transferable Adversarial Perturbations Using Region Proposal Networks and Beyond", *Computer Vision and Image Understanding (CVIU)*, Elsevier, vol. 213, 2021.
26. L. Wen†, D. Du†, Z. Cai, Z. Lei, M.-C. Chang, H. Qi, J. Lim, M.-H. Yang, S. Lyu*, "UA-DETRAC: A New Benchmark and Protocol for Multi-Object Detection and Tracking", *Computer Vision and Image Understanding (CVIU)*, Elsevier, 193, 2020.

27. M.-C. Chang, Y. Wei†, W.-R. Chen, C. Do*, "Deep Learning Based Super-Resolution for Small-Angle Neutron Scattering Data: Attempt to Accelerate Experimental Workflow", *MRS Communications*, 10 (1), 11-17, Mar. 2020.
28. M.-C. Chang, L. Ke†, H. Qi, L. Wen†, S. Lyu*, "Fast Online Video Pose Estimation by Dynamic Bayesian Modeling of Mode Transitions", *IEEE Trans. On Cybernetics*, pp. 1-14, 2019.
29. X. Wang†, M.-C. Chang, S. Lyu*, "Efficient Algorithms for Graph Regularized PLSA", *Pattern Recognition*, 2019.
30. M.-C. Chang*, T. Yu, J. Luo, K. Duan, P. Tu, Y. Zhao, N. Nagraj, V. Rajiv, M. Priebe, E. Wood, and M. Stachura, "Multi-Modal Sensor System for Pressure Ulcer Wound Assessment and Care", *IEEE Transactions on Industrial Informatics*, special section on multisensor fusion and integration for intelligent systems, Vol. 14, Issue 3, Mar. 2018.
31. L. Wen†, Z. Lei, M.-C. Chang, H. Qi*, and S. Lyu, "Multi-Camera Multi-Target Tracking with Space-Time-View Hypergraph", International Journal of Computer Vision (IJCV) Special Issue, Sep. 2016.
32. F. M. Fennessy*, A. Fedorov, T. Penzkofer, K. W. Kim, M. S. Hirsch, M. G. Vangel, P. Masry, T. A. Flood, M.-C. Chang, C. M. Tempany, R. V. Mulkern, S. N. Gupta, "Quantitative pharmacokinetic analysis of prostate cancer DCE-MRI at 3T: Comparison of two arterial input functions on cancer detection with digitized whole mount histopathological validation", *Magnetic Resonance Imaging*, Vol. 33, Issue 7, pp. 886-894, Elsevier, 2015.
33. *W. Huang and X. Li and Y. Chen and X. Li and M.-C. Chang *et al.*, "Variations of Dynamic Contrast-Enhanced Magnetic Resonance Imaging in Evaluation of Breast Cancer Therapy Response: A Multicenter Data Analysis Challenge", *Translational Oncology*, Vol. 7, n. 1, pp. 153-166, Feb. 2014.
34. M.-C. Chang* and B. B. Kimia, "Measuring 3D Shape Similarity by Graph-Based Matching of the Medial Scaffolds", *Computer Vision and Image Understanding (CVIU)*, Special Issue on 3D Imaging and Modeling (3DIM), Elsevier, Vol. 115, Issue 5, pp. 707-720, May. May 2011.
35. M.-C. Chang*, F. F. Leymarie, and B. B. Kimia, "Surface Reconstruction from Point Clouds by Transforming the Medial Scaffold", *Computer Vision and Image Understanding (CVIU)*, Elsevier, Vol. 113, Issue 11, pp. 1130-1146, Nov. 2009. (Top 5 downloaded computer science articles in ScienceDirect during April-June 2009.)
36. M.-C. Chang, C.-S. Fuh*, and H.-Y. Chen, "Fast Search Algorithms for Industrial Inspection", *International Journal of Pattern Recognition and Artificial Intelligence (IJPRAI)*, World Scientific, Vol. 15, No. 4, pp. 675-690, Jun. 2001.

Book / Book Chapter

- Xin Wang, Hui Guo†, Shu Hu, M.-C. Chang, Siwei Lyu, "GAN-Generated Faces Detection: A Survey and New Perspectives", Frontiers in Artificial Intelligence and Applications, *IOS Press Ebooks, ISBN: 9781643684369, Vol. 372, ECAI 2023,* pp. 2533-2542, Sep. 28, Sep. 2023.
- M.-C. Chang, "ObjectWindows Programming: Principle and Practice", 700 pages in Chinese Language, InfoWide Publisher Co., Ltd., ISBN 957-726-104-3, March 1994.

Prestigious Peer-Reviewed Conference and Workshop Publications

* Peer-reviewed computer science conferences often have greater impact and visibility than journals. All listed conference papers have been, or will be, presented as talks or posters; these presentations are not separately listed under Talks.

1. Y.-K. Hsieh, J.-W. Hsieh, K.-C. Peng, X. Li, C.-C. Sun, Y.-C. Tseng, M.-C. Chang, "Foveal-Mamba: Inside-Out Ring Scanning with Recurrent Offset Prediction for Visual State Space Models," *Neural Information Processing Systems (NeurIPS)*, Sydney, Australia, Dec., 2026.
2. Y.-K. Hsieh, X. Li, C.-C. Sun, Y.-C. Tseng, M.-C. Chang, K.-C. Peng, J.-W. Hsieh, "Counting Without Scale: Scale-Consistent Error Correction for Crowd Counting," *Neural Information Processing Systems (NeurIPS)*, Sydney, Australia, Dec., 2026.
3. C.-F. Hsu, Y.-H. Chiou, M.-C. Chang, W.-C. Chen, "Continual Learning with Support Boundary Experience Blending," **Best Paper Award,** *IEEE Multimedia Information Processing and Retrieval (MIPR)*, Bangkok, Thailand, Aug. 2026.

4. F. Ye, X. Li, A. Yu†, M.-C. Chang, L. Chu, D. Wertheimer, "FlashSinkhorn: IO-Aware Entropic Optimal Transport on GPU", *International Conference on Machine Learning (ICML)*, Seoul, South Korea, July, 2026 (**Oral Presentation**, **0.7%** of all submissions).
5. Y.-K. Hsieh, J.-W. Hsieh, X. Li, M.-C. Chang, Y.-C. Tseng, K.-C. Peng, "Partial Ring Scan: Revisiting Scan Order in Vision State Space Models", *International Conference on Machine Learning (ICML)*, Seoul, South Korea, July, 2026.
6. W. Lu, A. Yu†, J. Li, Z. Zhang, F. Ye, M.-C. Chang, "FAVE: A Structured Benchmark for Fine-Grained Audio-Visual Temporal Evaluation in Multimodal LLMs", *Computer Vision and Pattern Recognition (CVPR)*, Denver, CO, USA, June, 2026.
7. J.-L. Li,, M.-C. Chang, W.-C. Chen, "How Bias Binds: Measuring Hidden Associations for Bias Control in Text-to-Image Compositions", *AAAI Conference on Artificial Intelligence (AAAI)*, Singapore, Jan., 2026.
8. H.-B. Tan, C.-Y. Fu, M.-C. Chang, W.-C. Chen, "Latent Orthogonal Perturbation: A Data Poisoning Approach to Prevent Unauthorized LoRA Fine-tuning", *International Conference on Advanced Video and Signal-based Surveillance (AVSS)*, **Best Paper Runner-Up**, Tainan, Taiwan, Aug. 2025. (top-2 out of 147 paper submissions).
9. J. Hu, C. Feng, S. Hu, M.-C. Chang, X. Li, X. Wu, X. Wang, "RLMiniStyler: Light-weight RL Style Agent for Arbitrary Sequential Neural Style Generation", *International Joint Conference on Artificial Intelligence (IJCAI)*, Montreal, Canada, Sep. 2025 (acceptance rate **19.3%**).
10. Z. Zhang†, M.-C. Chang, X. Li, "Training-Free Image Manipulation Localization with Diffusion Model", *AAAI Conference on Artificial Intelligence (AAAI)*, Philadelphia, PA, USA, 2025 (acceptance rate **23.4%**).
11. Z. Zhang, Y. Zhao, M.-C. Chang, C. Lin, J. Liu, "E4: Energy-Efficient DNN Inference for Edge Video Analytics Via Early Exiting and DVFS", *AAAI Conference on Artificial Intelligence (AAAI)*, Philadelphia, PA, USA, Apr. 2025 (acceptance rate **23.4%**).
12. Y. Zhang, J.-W. Hsieh, X. Li, M.-C. Chang, C.-C. Lee, K.-C. Fan, "MOTE-NAS: Multi-Objective Training-based Estimate for Efficient Neural Architecture Search", *Conference on Neural Information Processing Systems (NeurIPS)*, Vancouver, Canada, Dec., 2024 (acceptance rate **25.8%**).
13. J.-L. Li, M.-C. Chang, W.-C. Chen, "Learning Multi-Manifold Embedding for Out-Of-Distribution Detection", *Beyond Euclidean Workshop (BEW): Hyperbolic and Hyperspherical Learning for Computer Vision*, *European Conference on Computer Vision (ECCV) Workshop,* **Best Paper Award**, Milan, Italy, 2024 (out of 9 proceeding track and 9 extended abstract papers).
14. Z. Zhang†, M. Li, X. Li, M.-C. Chang, J.-W. Hsieh, "Image Manipulation Detection With Implicit Neural Representation and Limited Supervision", *European Conference on Computer Vision (ECCV)*, Milan, Italy, Nov. 2024 (acceptance rate **27.9%**).
15. Z. Zhang†, M. Li, M.-C. Chang, "A New Benchmark and Model for Challenging Image Manipulation Detection", *AAAI Conference on Artificial Intelligence (AAAI)*, Vancouver, Canada, Mar. 2024 (acceptance rate **23.75%**).
16. Y.-H. Wang, J.-W. Hsieh, P.-Y. Chen, M.-C. Chang, H.-H. So, X. Li, "SMILEtrack: SiMIlarity LEarning for Occlusion-Aware Multiple Object Tracking", *AAAI Conference on Artificial Intelligence (AAAI)*, Vancouver, Canada, 2024 (acceptance rate **23.75%**).
17. Y.-Y. Chen, J.-W. Hsieh, X. Li, M.-C. Chang, "Pushing the Limit of Fine-Tuning for Few-Shot Learning: Where Feature Reusing Meets Cross-Scale Attention", *AAAI Conference on Artificial Intelligence (AAAI)*, Vancouver, Canada, 2024 (acceptance rate **23.75%**).
18. Z. Zhang†, M.-C. Chang, "Two-Stage Dual Augmentation with CLIP for Improved Text-to-Sketch Synthesis", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, Singapore, 2023, **Best Student Paper Award** (out of 23 oral papers).
19. S. Weng, Y. Wei†, M.-C. Chang, B. Shi, "Instance Contour Adjustment via Structure-driven CNN", *European Conference on Computer Vision (ECCV)*, Tel Aviv, Israel, Nov. 2022 (acceptance rate **28%**).
20. B.-S. Wang, J.-W. Hsieh, P.-Y. Chen, M.-C. Chang, L. Ke, S. Lyu, "LDW-Pooling: Learning Discrete Wavelet Pooling for Convolutional Networks", *British Machine Vision Conference (BMVC),* 2021 (acceptance rate **33.6%**).

21. Y.-H. Huang, J.-W. Hsieh, M.-C. Chang, L. Ke, S. Lyu, A. S. Santra, "Multi-Teacher Single-Student Visual Transformer with Multi-Level Attention for Face Spoofing Detection", *British Machine Vision Conference (BMVC)*, 2021 (acceptance rate **33.6%**).
22. Y. Wei†, P. Zhang, W. Li†, Z. Gan, S. Lyu, M.-C. Chang, L. Zhang, J. Gao, "MagGAN: High-Resolution Face Attribute Editing with Mask-Guided Generative Adversarial Network'", *Asian Conference on Computer Vision (ACCV)*, 2020 (acceptance rate **28%**).
23. Y. Wei†, W. Li†, Y. Fan†, L. Xu, M.-C. Chang, S. Lyu, "3D Single-Person Concurrent Activity Detection Using Stacked Relation Network", *AAAI Conference on Artificial Intelligence (AAAI)*, New York, NY, USA, Apr. 2020. (Oral Presentation, acceptance rate **20.6%**)
24. Y. Li†, L. Wen†, M.-C. Chang and S. Lyu, "Graph-to-Graph Energy Minimization for Video Object Segmentation", International Conference on Advanced Video and Signal-based Surveillance (AVSS), Taipei, Taiwan, Sep. 2019. (Oral presentation, acceptance rate **16%**)
25. Y. Li†, X. Bian, M.-C. Chang, S. Lyu, "Exploring the Vulnerability of Single Shot Module in Object Detectors via Imperceptible Background Patches", *British Machine Vision Conference (BMVC)*, Cardiff, UK, 2019. (Acceptance rate **28%**)
26. L. Ke†, M.-C. Chang, H. Qi, S. Lyu, "Multi-Scale Structure-Aware Network for Human Pose Estimation", *European Conference on Computer Vision (ECCV)*, Munich, Germany, 2018. (Acceptance rate **31.8%**)
27. Y. Li†, D. Tian, M.-C. Chang, X. Bian, S. Lyu, "Universal Adversarial Perturbation on Deep Proposal-based Models", *British Machine Vision Conference (BMVC)*, Newcastle, UK, 2018. (Acceptance rate **29.9%**)
28. W. Li†, L. Wen†, M.-C. Chang, S. Lim, and S. Lyu, "Adaptive RNN Tree for Large-Scale Human Action Recognition", *IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, Oct. 2017. (Acceptance rate **29%**)
29. X. Wang†, M.-C. Chang, Y. Ying†, and S. Lyu, "Co-regularized PLSA for Multi-Modal Learning", *AAAI Conference on Artificial Intelligence (AAAI),* Phoenix, AR, USA, Mar. 20166. (Acceptance rate **26%**)
30. M.-C. Chang, H. Qi, X. Wang†, H. Cheng, and S. Lyu, "Fast Online Upper Body Pose Estimation from Video", *British Machine Vision Conference (BMVC)*, Swansea, UK, 2015. (Acceptance rate **33%**)
31. Y. Kim, J. Chen, M.-C. Chang, E. M. Provost, X. Wang, and S. Lyu, "Joint Event Localization and Classification of Human Action Videos with Event Transitions", *IEEE Automatic Face and Gesture Recognition (FG)*, Oral presentation, Ljubljana, Slovenia, 2015. (Acceptance rate **12%**)
32. Y. Zhang, X. Liu, M.-C. Chang, W. Ge, and T. Chen, "Spatio-Temporal Phrases for Activity Recognition", *in Proceeding of European Conference on Computer Vision (ECCV),* Firenze, Italy, Oct. 2012. (Acceptance rate **25%**)
33. Y. Zhang, W. Ge, M.-C. Chang, and X. Liu, "Group-Level Context Learning for Event Recognition", *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV),* Oral presentation, **Best Student Paper Award** (top 2 out of 63 accepted papers), Breckenridge, Colorado, USA, Jan. 2012.
34. M.-C. Chang, N. Krahnstoever, and W. Ge, "Probabilistic Group-Level Motion Analysis and Scenario Recognition", *IEEE International Conference on Computer Vision (ICCV)*, Barcelona, Spain, Nov. 2011. (Acceptance rate **23.7%**)
35. N. Krahnstoever, M.-C. Chang, and W. Ge, "Gaze and Body Pose Estimation from a Distance", *IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS)*, **Best Paper Award - Runner up** (top 2 out of 81 accepted papers), Klagenfurt, Austria, Aug. - Sep. 2011.
36. M.-C. Chang and B. B. Kimia, "Regularizing 3D Medial Axis Using Medial Scaffold Transforms", *IEEE Computer Vision and Pattern Recognition (CVPR)*, Anchorage, Alaska, USA, pp. 1-8, Jun. 2008. (Acceptance rate **32%**)

Additional Peer-Reviewed Conference and Workshop Publications

1. S. Dewan, H. Elgala, A. Tzadok, M.-C. Chang, "Jointly Learned Waveforms for Residual CFO Robustness and PAPR Reduction", *IEEE Military Communications Conference (MILCOM)*, Washington DC, USA, Oct. 2026.
2. A. K. Pandey†, M.-C. Chang, J.-W. Hsieh, "TRACE-TAR: Reusable Video Evidence and Shared Event Memory for Transportation Anomaly Reasoning", *AI City Challenge Workshop at European Conference on Computer Vision (ECCV)*, Malmö, Sweden, Sep. 2026.

3. T.-Y. Tsai*†, A. Yu*, L. Lee, F. X.-F. Ye, D. S. Shin, T.-J. Kao, X. Li, M.-C. Chang, "RatSeizure: A Benchmark and Saliency-Context Transformer for Rat Seizure Localization", *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Strasbourg, France, Sep. 2026.
4. A. K. Pandey†, M.-C. Chang, "ForensiText: ROI-Guided Multimodal Forensics for Scene Text Tampering Detection and Localization", *IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS),* Lecce, Italy, Aug. 2026.
5. N. Xu, M.-C. Chang, "Detecting ATM Money Mule Activities via LVLM-Guided Forensic Analysis," *IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS)*, Lecce, Italy, Aug. 2026.
6. P.-T. Ku, J.-W. Hsieh, Y.-K. Hsieh, Y.-T. Hsieh, C.-C. Sun, M.-C. Chang, X. Li, "MAESTRO: Multi-Anchor and Dual-Constraint Enhanced Sequential Training for Robust Optimization in Image Editing", *IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS)*, Lecce, Italy, Aug. 2026.
7. C.-F. Hsu, Y.-H. Chiou, P.H. Huang, Y.-C. Liao, S.-H. Yang, J.-Z. Tang, Y.-L. Chang, M.-C. Chang, "Bridging Design and Quality: A Technical Overview of IFMs in Manufacturing", IEEE Multimedia Information Processing and Retrieval (MIPR), Bangkok, Thailand, Aug. 2026.
8. C.-M. Tsai, J.-W. Hsieh, M.-C. Chang, "BF-YOLOv7: Enhancing Helmet Rule Violation Detection", *Advances and Trends in Artificial Intelligence. Theory and Applications (IEA/AIE)*, Kitakyushu, Japan, July 2025.
9. Y.-T. Hsieh, C.-C. Hsu, X. Li, M.-C. Chang, J.-W. Hsieh, "SILVA-Mamba: Spatial-Integrity For Landslide Segmentation Via Vectorized HILBERT Scanning and Adaptive Mamba", *IEEE International Conference on Image Processing (ICIP)*, Tampere, Finland, Sep. 2026.
10. W. Lu, J. Li, A. Yu†, M.-C. Chang, "LLaVA-MR: Large Language-and-Vision Assistant for Video Moment Retrieval", *International Conference on Pattern Recognition (ICPR),* Lyon, France, Aug. 2026.
11. C.-P. Chen, J.-L. Li, M.-C. Chang, "Uncertainty-Aware Proxy Attribute Reasoning for Reliable Media Bias Detection", *ACL TrutNLP Workshop*, San Diego, CA, USA, July 2026.
12. M.-C. Chang, T.-Yu Tsai†, D. Wertheimer, F. Ye, "Causal Volterra Dynamics of Mamba", International Conference on Machine Learning (ICML) Workshop on High-dimensional Learning Dynamics (HiLD), 2026.
13. C.-M. Lee†, Y.-F. Lin†, W.-H. Tsai, M.-C. Chang, X. Li, C.-C. Hsu, "FracQuant: Rank-Adaptive Image Super-resolution Network Quantization via Fractal Complexity Assessment", *International Conference on Machine Learning (ICML) CoLoRAI Workshop*, Seoul, South Korea, July, 2026.
14. R. Wang†, M.-C. Chang, E. Lackman-Westin, "PreClaim-GCT: Self-Supervised Graph–Transformer Pretraining for Incident Outcome Prediction from Healthcare Claims", *Artificial Intelligence in Medicine (AIME)*, Ottawa, Canada, Jul. 2026.
15. J.-W. Hsieh, Y.-H. Wu, Y.-K. Hsieh, X. Li, K.-C. Peng, M.-C. Chang, "SoREL: Soft-Label Refurbishment with Ensemble Learning for Noisy Long-Tailed Classification", *Computer Vision and Pattern Recognition (CVPR) Findings Workshop*, Denver, CO, USA, June Jun. 2026.
16. C.-M. Tsai, J.-W. Hsieh, M.-C. Chang, "Optimization-Diverse Transformer Adaptation with Gated Heterogeneous Fusion for Robust Thermal Object Detection", *Computer Vision and Pattern Recognition (CVPR) MaCVi Workshop*, Denver, CO, USA, June 2026.
17. P.-H. Huang, J.-L. Li, P.-H. Huang, M.-C. Chang, W.-C. Chen, "PatchEAD: Unifying Industrial Visual Prompting Frameworks for Patch-Exclusive Anomaly Detection", *Winter Conference on Applications of Computer Vision (WACV)*, Tuscon, AZ, USA, Mar. 2026.
18. S. R. Rajan†, M.-C. Chang, P. K. Atrey, "Physics-Guided Exposure Parameter Estimation for Image Metadata Verification", International Symposium on Multimedia (ISM), Naples, Italy, Nov. 2025.
19. J.-Z. Tang, C.-F. Hsu, J.-L. Li, M.-C. Chang, W.-C. Chen, "LSSGen: Leveraging Latent Space Scaling in Flow and Diffusion for Efficient Text to Image Generation', *ICCV AIGENS Workshop*, Honolulu, HI, USA, Oct. 2025.
20. Z. Tang, S. Wang, D. Anastasiu, M.-C. Chang, A. Sharma, Q. Kong, N. Kobori, M. Gochoo, G. Batnasan, M.E. Otgonbold, F. Alnajjar, J.-W. Hsieh, T. Kormuta, X. Li, Y. Zhao, H. Zhang, S. Radhakrishnan, A. Jain, R. Kumar, V. Murali, Y. Wang, S. S. Pusegaonkar, Y. Wang, S. Biswas, X. Wu, Z. Zheng, P. Chakraborty, R. Chellappa, "The 9th AI City Challenge", *ICCV AI City Challenge Workshop*, Honolulu, HI, USA, Oct. 2025.
21. T.-Y. Tsai†, A. Yu†, Wenqi Li, M.-C. Chang, "Latent-Guided Hybrid Architecture for Liver Segmentation in Contrast-Enhanced MRI", *MICCAI CARE Challenge*, Daejeon, Korea, 2025.

22. Y.-K. Hsieh, J.-W. Hsieh, X. Li, Y.-M. Zhang, Y.-C. Tseng, M.-C. Chang, "Scale-Aware Crowd Counting Network with Annotation Error Modeling", *IEEE International Conference on Image Processing (ICIP)*, Anchorage, AK, USA, Sep., 2025.
23. H. B. Tan, C.-Y. Fu, M.-C. Chang, W.-C. Chen, "Latent Orthogonal Perturbation: A Data Poisoning Approach to Prevent Unauthorized LoRA Fine-Tuning", *IEEE International Conference on Advanced Visual and Signal-Based Systems (AVSS)*, Tainan, Taiwan, Aug. 2025.
24. T.-L. Yang, J.-S. Liu, Y.-H. Tseng, J.-S. Jang, M.-C. Chang, W.-C. Chen, "Enhancing LLM Question Answering with RAG through Dense Vector Search and Re-Ranking", *IEEE International Conference on Advanced Visual and Signal-Based Systems (AVSS)*, Aug. 2025.
25. T. Y. Tsai†, A. Yu†, M. S. Maadugundu, I. J. Mohima, U. H. Barsha, M.-H. F. Chen, B. Prabhakaran, M.-C. Chang, "IntelliCardiac: An Intelligent Platform for Cardiac Image Segmentation and Classification", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, Aug. 2025.
26. C.-C. Hsu, C.-M. Lee, M.-T. Ko, E.-C. Liu, Y.-C. Cheng, M.-C. Chang, "HyperForensics++: Toward Adversarial Perturbed and Object Replacement in Hyperspectral Imaging Dataset", *IEEE International Geoscience and Remote Sensing Symposium (IGRASS)*, Brisbane, Australia, 2025.
27. Y.-T. Ku, M.-C. Ho, Y. Xiao, F.-H. Liu, C.-F. Hsu, M.-C. Chang, S.-H. Hung, W.-C. Chen, "A Comprehensive Evaluation of Encrypted DNN Inference Methods", *International Symposium on Circuits and Systems (ISCAS)*, London, UK, May 2025.
28. Y. Chen†, M.-C. Chang, M. Kirchner, Z. Zhang, X. Li, A. Basharat, A. Hoogs, "A Semantically Impactful Image Manipulation Dataset: Characterizing Image Manipulations Using Semantic Significance", *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Tucson, AZ, USA, Feb. 2025.
29. P.-H. Huang, J.-L. Li, C.-P. Chen, M.-C. Chang, W.-C. Chen, "Who Brings the Frisbee: Probing Hidden Hallucination Factors in Large Vision-Language Model via Causality Analysis", *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Tucson, AZ, USA, Feb. 2025.
30. C.-Y. Lin, B.-H. Lai, H.-F. Ng, W.-Y. Lin, M.-C. Chang, "Robust Defense Against Adversarial Attacks with Defensive Preprocessing and Adversarial Training", *IEEE International Conference on Consumer Electronics (ICCE) in conjunction with CES*, Las Vegas, NV, USA, Jan. 2025.
31. M.-C. Ho, Y.-T. Ku , Y. Xiao, F.-H. Liu, C.-F. Hsu, M.-C. Chang, S.-H. Hung, W.-C. Chen, "Efficient Design of FHEW/TFHE Bootstrapping Implementation with Scalable Parameters", *IEEE/ACM International Conference on Computer Aided Design*, Newark, NJ, USA, Oct. 2024.
32. Y.-M. Zhang, Y.-H. Chang, J.-W. Hsieh, X. Li, M.-C. Chang, C.-C. Lee, K.-C. Fan, "SET-NAS: Sample-Efficient Training for Neural Architecture Search with Strong Predictor and Stratified Sampling", *IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, UAE, 2024.
33. P.-H. Huang, C.-C. Lin, C.-F. Hsu, M.-C. Chang, W.-C. Chen, "Learning with Instance-Dependent Noisy Labels by Anchor Hallucination and Hard Sample Label Correction", *IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, UAE, Oct. 2024.
34. A. Pandey†, M.-C. Chang, X. Li, "TextSleuth: A New Dataset and Baseline for Scene Text Manipulation Detection and Localization", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, 2024.
35. Y. Chen†, M.-C. Chang, X. Li, "Leveraging Semantic Segmentation for Image Manipulation Detection and Localization", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, Aug. 2024.
36. Z. Zhang†, T.-W. Huang, G.-M. Su, M.-C. Chang, X. Li, "Diffusion Model for Text to Synchronized Joint Talking Head Video and Audio Generation", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, 2024.
37. T.-Y. Tsai†, L. Lin, S. Hu, M.-C. Chang, H. Zhu, X. Wang, "UU-Mamba: Uncertainty-aware U-Mamba for Cardiac Image Segmentation", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, 2024.
38. A. Yu†, J. Varghese, F. Demirkiran, P. Buonaiuto, X. Li, M.-C. Chang, "Dual-Phase MSQNet for Species-Specific Animal Activity Recognition", *IEEE International Conference on Multimedia and Expo (ICME),*

Grant Challenge on Multi-Modal Video Reasoning and Analyzing Competition (MMVRAC) Animal Action Recognition, **2nd Place Winner**, Niagara Falls, ON, Canada, Jul. 2024.

39. S. Wang, D. C. Anastasiu, Z. Tang, M.-C. Chang, Y. Yao, L. Zheng, M. S. Rahman, M. Arya, N. Kobori, M. Gochoo, M.-E. Otgonbold, F. Alnajjar, G. Batnasan, P.-Y. Chen, J.-W. Hsieh, X. Wu, S. Pusegaonkar, Y. Wang, S. Biswas, R. Chellappa, "The 8th AI City Challenge", *IEEE CVPR Workshop on the AI City Challenge*, Seattle, WA, USA, 2024.
40. S.-J. Kuo, P.-H. Huang, C.-C. Lin, J.-L. Li, M.-C. Chang, "Improving Limited Supervised Foot Ulcer Segmentation Using Cross-Domain Augmentation Strategies", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Seoul, Korea, Apr. 2024.
41. P.-Y. Chen, J.-W. Hsieh, M.-C. Chang, M. Gochoo, F.-P. Lin, Y.-S. Chen, "Fisheye Multiple Object Tracking by Learning Distortions without Dewarping", *IEEE International Conference on Image Processing (ICIP)*, Kuala Lumpur, Malaysia, Oct. 2023.
42. T.-L. Liu, Y.-T. Ku, M.-C. Ho, F.-H. Liu, M.-C. Chang, C.-F. Hsu, W.-C. Chen and S.-H. Hung, "An Efficient CKKS-FHEW/TFHE Hybrid Encrypted Inference Framework", *International Workshop on Private, Secure, and Trustworthy AI (PriST-AI) in Conjunction with ESORICS*, 2023.
43. X. Wang, H. Guo†, S. Hu, M.-C. Chang, Siwei Lyu, "GAN-generated Faces Detection: A Survey and New Perspectives", *European Conference on Artificial Intelligence (ECAI)*, Krakaw, Poland, Sep. 2023.
44. Y. Chen†, M.-C. Chang, "Transformer Based Image-Text Consistency Analysis for Infographic Articles", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, Singapore, 2023.
45. K.-H. Li, C.-F. Hsu, M.-C. Chang, F.-H. Liu, S.-Y. Chien, W.-C. Chen, "Region-aware Photo Assurance System for Image Authentication", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, Singapore, 2023.
46. P.-Y. Chen, J.-W. Hsieh, M.-C. Chang, C.-I Huang, W.-Y. Chen, M. Gochoo, F.-P. Lin, "Fisheye-DETRAC: A New Fisheye Video Benchmark for Multi-Object Detection and Tracking", *IPPR Computer Vision, Graphics, and Image Processing (CVGIP)*, Kinmen, Taiwan, 2023.
47. M. Naphade, S. Wang, D. C. Anastasiu, Z. Tang, M.-C. Chang, Y. Yao, L. Zheng, M. S. Rahman, A. Venkatachalapathy, A. Sharma, Q. Feng, V. Ablavsky, S. Sclaroff, P. Chakraborty, A. Li, S. Li, R. Chellappa, "The 7th AI City Challenge", *IEEE CVPR Workshop on the AI City Challenge*, Vancouver, Canada, 2023.
48. B.-S. Wang, P.-Y. Chen, Y.-K. Hsieh, J.-W. Hsieh, M.-C. Chang, J. He, S.-Y. Teng, H. Yue, "PRB-FPN+: Video Analytics for Enforcing Motorcycle Helmet Laws", *IEEE CVPR Workshop on the AI City Challenge*, Vancouver, Canada, 2023.
49. C.-M. Tsai, J.-W. Hsieh, M.-C. Chang, G.-L. He, P.-Y. Chen, W.-T. Chang, Y.-K. Hsieh, "Video Analytics for Detecting Motorcyclist Helmet Rule Violations", *IEEE CVPR Workshop on the AI City Challenge*, Vancouver, Canada, 2023.
50. M. Gochoo, M.-E. Otgonbold, E. Ganbold, J.-W. Hsieh, M.-C. Chang, P.-Y. Chen, B. Dorj, H. A. Jassmi, G. Batnasan, F. Alnajjar, M. Abduljabbar, F.-P. Lin, "FishEye8K: A Benchmark and Dataset for Fisheye Camera Object Detection", *IEEE CVPR Workshop on the AI City Challenge*, Vancouver, Canada, Jun. 2023.
51. F. Akhyar, L. Novamizanti, M. Azka I., I. H. P.ratama, S. R. Firmansyach, M.-C. Chang, C.-Y. Lin, "Observation of Attention Mechanism Baseline for PCB Surface Inspection System", *IEEE Asia Pacific Conference on Wireless and Mobile (APWiMob)*, 2022.
52. E. AlBadawy†, C. Sun, S. Robinson, T. Davison, M.-C. Chang, S. Lyu, "Exposing AI-Synthesized Human Voices Using Neural Vocoder Artifacts", *IEEE International Workshop on Information Forensics and Security (WIFS)*, 2022
53. Y. Chen† and M.-C. Chang, "Toward Multimodal Semantic Consistency Analysis of Long Form Articles", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2022.
54. C.-F. Hsu, M.-C. Chang, W.-C. Chen, "A Robust Collaborative Learning Framework Using Data Digests and Synonyms to Represent Absent Clients", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, Aug. 2022.
55. C.-F. Hsu, J.-L. Huang, F.-H. Liu, M.-C. Chang, W.-C. Chen, "FedTrust: Towards Building Secure Robust and Trustworthy Moderators for Federated Learning", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2022.

56. F. Akhyar, L. Novamizanti, T. Putra, E. N. Furqon, M.-C. Chang, C.-Y. Lin, "Lightning YOLOv4 for a Surface Defect Detection System for Sawn Lumber", *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, Aug. 2022.
57. H. Guo†, S. Hu, M.-C. Chang, S. Lyu, "Open-Eye: An Open Platform to Study Human Performance on Identifying AI-Synthesized Faces", Demo Paper, *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2022.
58. M. Naphade, S. Wang, D. Anastasiu, Z. Tang, M.-C. Chang, Y. Yao, L. Zheng, M. Rahman, A. Venkatachalapathy, A. Sharma, Q. Feng, V. Ablavsky, S. Sclaroff, P. Chakraborty, A. Li, S. Li, R. Chellappa, "The 6th AI City Challenge", *CVPR Workshop on AI City Challenge*, New Orleans, LA, 2022.
59. S. Hu†, C.-H. Liu, J. Dutta, M.-C. Chang, S. Lyu, N. Ramakrishnan, "PseudoProp: Robust Pseudo-Label Generation for Semi-Supervised Object Detection in Autonomous Driving Systems", *CVPR Workshop on Autonomous Driving (WAD),* New Orleans, LA, 2022.
60. E. AlBadawy†, A. Gibiansky, Q. He, J. Wu, M.-C. Chang, S. Lyu, "VocBench: A Neural Vocoder Benchmark for Speech Synthesis", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore, 2022.
61. H. Guo†, S. Hu†, X. Wang†, M.-C. Chang, S. Lyu, "Eyes Tell All: Irregular Pupil Shapes Reveal GAN-generated Faces", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore, May 2022.
62. Z. Zhang†, M.-C. Chang, T. Bui, "Improving Class Activation Map for Weakly Supervised Object Localization", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore, May 2022.
63. M. Huang, S. Jia, M.-C. Chang, S. Lyu, "Text-Image De-contextualization Detection Using Vision-Language Models", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore, 2022.
64. S.-F. Wu, M.-C. Chang, S. Lyu, C.-S. Wong, A. Pandey†, P.-C. Su, "FlagDetSeg: Multi-Nation Flag Detection and Segmentation in the Wild", *IEEE Advanced Video and Signal-based Surveillance (AVSS)*, Nov. 2021.
65. G. Zhao, A. Pandey†, M.-C. Chang, S. Lyu, "A Video Analytic System for Rail Crossing Point Protection", *IEEE Advanced Video and Signal-based Surveillance (AVSS)*, Nov. 2021.
66. P.-Y. Chen, M.-C. Chang, J.-W. Hsieh, Y.-S. Chen, B.-S. Wang, C.-H. Chou, B.-C. Lin, "Effective LightWeight Feature Fusion Feature Pyramid Network for Extremely Small Object Detection", *IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP)*, Taiwan, 2021.
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8. Y.-T. Ku, Y. Xiao, M.-C. Ho, C.-F. Hsu, W.-C. Chen, F.-H. Liu, M.-C. Chang, S.-H. Hung, "Method for Machine Learning Model Verification and Transaction," US Patent App. 18/420,744, 2025.
9. C.-F. Hsu, W.-C. Chen, J.-L. Huang, M.-C. Chang, F.-H. Liu, "Method for Verifying Model Update," US Patent 12,316,737, 2025.
10. C.-F. Hsu, W.-C. Chen, M.-C. Chang, "Method of Training a Model Using Data Essence," US Patent App. 18/665,405, 2025.
11. J.-L. Li, W.-C. Chen, M.-C. Chang, "Multi-Manifold Embedding Learning Method and System," US Patent App. 18/415,620, 2025.
12. T.-L. Liu, Y.-T. Ku, M.-C. Ho, C.-F. Hsu, W.-C. Chen, F.-H. Liu, M.-C. Chang, S.-H. Hung, "Operating System and Method of a Fully Homomorphic Encryption Neural Network Model," US Patent App. 18/545,775, 2025.
13. J.-L. Li, W.-C. Chen, M.-C. Chang, "Robust Out-of-Distribution Detection System and Method," US Patent App. 18/665,267, 2025.
14. S.-J. Kuo, P.-H. Huang, C.-C. Lin, J.-L. Li, M.-C. Chang, W.-C. Chen, "Segmentation Model Training Method, Device, and Non-Transitory Computer Readable Storage Medium," US Patent App. 18/418,329, 2025.
15. K.-H. Li, C.-F. Hsu, W.-C. Chen, M.-C. Chang, F.-H. Liu, "System for Verifying Edited Image," US Patent 12,430,472, 2025.
16. C.-F. Hsu, W.-C. Chen, M.-C. Chang, "Federated Learning System and Method of Protecting Data Digest," US Patent App. 18/205,522, 2024.
17. Y.-T. Ku, C.-F. Hsu, W.-C. Chen, F.-H. Liu, M.-C. Chang, "Privacy Computing Method Based on Homomorphic Encryption," US Patent App. 18/171,319, 2024.
18. C.-F. Hsu, W.-C. Chen, M.-C. Chang, "Federated Learning Method Using Synonym," US Patent App. 17/699,263, 2023.
19. C.-F. Hsu, W.-C. Chen, M.-C. Chang, "Federated Learning System and Method Using Data Digest," US Patent App. 18/084,810, 2023.
20. M.-C. Chang, J. Ping, E. M. Gros, A. Jain, P. Tu, "Image and Video Capture Architecture for Three-Dimensional Reconstruction," US Patent 10,600,194, 2020.
21. E. M. Gros, J. Ping, A. Jain, M.-C. Chang, P. Tu, "Three-Dimensional Modeling of an Object," US Patent 10,679,338, 2020.
22. J. Chen, P. Tu, M.-C. Chang, Y. Kim, S. Lyu, "Systems and Methods for Analyzing Time Series Data Based on Event Transitions," US Patent 9,984,154, 2018.
23. Y. Yao, P. Tu, M.-C. Chang, L. Guan, Y. Tong, "Apparatus and Method for Predicting Solar Irradiance Variation," US Patent 8,923,567, 2014.
24. L. Guan, P. Tu, Y. Yao, M.-C. Chang, "Apparatus and Method for Spatially Relating Views of Sky Images Acquired at Spaced Apart Locations," US Patent 8,750,566, 2014.
25. P. Tu, M. L. Grabb, X. Liu, T. Yu, Y. Yao, D. Gao, M.-C. Chang, "Analytics-to-Content Interface for Interactive Advertising," US Patent App. 13/308,376, 2013.
26. N. Krahnstoever, P. Tu, M.-C. Chang, W. Ge, "Person Tracking and Interactive Advertising," US Patent App. 13/221,896, 2013.
27. P. Tu, M. L. Grabb, X. Liu, T. Yu, Y. Yao, D. Gao, M.-C. Chang, "Usage Measurement Techniques and Systems for Interactive Advertising," US Patent App. 13/308,386, 2013.

Taiwan Patents

28. Y.-T. Ku, Y. Xiao, M.-C. Ho, C.-F. Hsu, W.-C. Chen, F.-H. Liu, M.-C. Chang, S.-H. Hung, "Method for Machine Learning Model Verification and Transaction," Taiwan Patent TWI875327B, 2025.

29. J.-L. Li, W.-C. Chen, M.-C. Chang, "Multi-Manifold Embedding Learning Method and System," Taiwan Patent TWI854880B, 2024.

China Patents

30. Y.-T. Ku, Y. Xiao, M.-C. Ho, C.-F. Hsu, W.-C. Chen, F.-H. Liu, M.-C. Chang, S.-H. Hung, "Method for Machine Learning Model Verification and Transaction," China Patent Publication CN120013531A, 2025.
31. N. Krahnstoever, P. Tu, M.-C. Chang, W. Ge, "Person Tracking and Interactive Advertising," China Patent Publication CN102982753A, 2013.

Patent Disclosures

2024

1. NP-35565-US/230300.US, S.-J. Kuo, P.-H. Huang, C.-C. Lin, J.-L. Li, M.-C. Chang, W.-C. Chen, "Segmentation Model Training Method, Device, and Non-Transitory Computer Readable Storage Medium," Jan. 2024.
2. NP-35583-US/230304.US, Y.-T. Ku, Y. Xiao, M.-C. Ho, C.-F. Hsu, W.-C. Chen, F.-H. Liu, M.-C. Chang, S.-H. Hung, "Method for Machine Learning Model Verification and Transaction," Jan. 2024.
3. NP-35229-US/230306.US, J.-L. Li, W.-C. Chen, M.-C. Chang, "Multi-Manifold Embedding Learning Method and System," Jan. 2024.

2022

4. IPD220 .US-TE2111018 (111US17348), C.-F. Hsu, M.-C. Chang, W.-C. Chen, "Federated Learning Method Using Synonym," Mar. 2022.
5. IPD220 .US-TE2201001 (111US17349), C.-F. Hsu, M.-C. Chang, W.-C. Chen, "Federated Learning System Using Synonym," Mar. 2022.

2017

6. GE LC:0111, M.-C. Chang, J. Ping, E. M. Gros, A. Jain, P. Tu, "Image and Video Capture Architecture for Three-Dimensional Reconstruction," Aug. 2017.
7. GE Disclosure 314571-1 LC:0110, E. M. Gros, J. Ping, A. Jain, M.-C. Chang, P. Tu, "Three-Dimensional Modeling of an Object," Aug. 2017.

2016

8. GE Disclosure, T. Yu, M.-C. Chang, Jiajia Luo, K. Duan, B. Hathaway, P. Tu, V. Rajiv, N. Nagraj, "A Multi-modality Portable Scanner for Wound Assessment and Care," Aug. 2016.
9. GE Disclosure, T. Yu, M.-C. Chang, Jiajia Luo, K. Duan, B. Hathaway, P. Tu, "A Deep Learning Based Health Care System for Patient Activity Monitoring," Aug. 2016.
10. GE Disclosure 69272, A. Can, M.-C. Chang, Jiajia Luo, "Automated Defect Recognition Using Sparsity Based Normalcy Modeling," Jan. 8, 2016.

2015

11. GE Disclosure 67929 285274, M.-C. Chang, J. Chen, P. Tu, "A Live Video Analytic System for Affect Analysis in Public Space," May 2015.

2014

12. GE Disclosure 58308 277205, Y. Kim, J. Chen, M.-C. Chang, S. Lyu, P. Tu, "Joint Localization and Classification of Events for Time Series Data by Explicitly Modeling Event Transitions," US patent in application, Jul. 2014.

2011

13. GE Disclosure 34920 256289, Guan Li, Y. Yao, M.-C. Chang, P. Tu, "Calibration-Free Multiple Sky Imager Image Alignment and View Interpolation," Oct. 2011.
14. GE Disclosure 34176 254698, X. Liu, M.-C. Chang, W. Ge, Y. Zhang, "Co-occurring Patterns Discovery from One-Dimensional Signals," Aug. 2011.

15. GE Disclosure 28430 251344, N. Krahnstoever, M.-C. Chang, W. Ge, P. Tu, "Remote Estimation of Gaze for Interactive Advertising," US and China patents in application, May 2011.
16. GE Disclosure 29689 251009, P. Tu, M. L. Grabb, N. Krahnstoever, X. Liu, T. Yu, Y. Yao, D. Gao, M.-C. Chang, "Usage Measurement Methods for Interactive Advertising," May 2011.
17. GE Disclosure 29681 251010, M. L. Grabb, P. Tu, N. Krahnstoever, X. Liu, T. Yu, Y. Yao, D. Gao, M.-C. Chang, "Episodic Approaches to Interactive Advertising," May 2011.
18. GE Disclosure 29659 250984, M. L. Grabb, P. Tu, N. Krahnstoever, X. Liu, T. Yu, Y. Yao, D. Gao, M.-C. Chang, "An Analytics-to-Content Interface for Interactive Advertising Applications," May 2011.
19. GE Disclosure 28509 250394, L. Guan, Y. Yao, M.-C. Chang, P. Tu, M. Schmidt, M. Lynass, "Optimal Optical Sensor Positioning for Cloud-Sun Occlusion Prediction at Solar Energy Plant," Feb. 2011.
20. GE Disclosure 28431 249978, Y. Yao, M.-C. Chang, L. Guan, Y. Tong, P. Tu, "Cloud Tracking for Solar Irradiance Prediction," Feb. 2011.

Public Talks, Poster Sessions, Demos

1. Invited seminar, "Agentic and Trustworthy AI for Multimodal World-Aware Systems", IEEE Computational Intelligence Society, 2026.
2. Invited seminar, "From Early AI to Agentic and Trustworthy AI Systems", *Formosa Association of Student Cultural Ambassadors (FASCA) Training on AI Literacy and Innovation*, Apr., 2026.
3. **Keynote**, "Trustworthy AI in a Smarter World: Addressing Awareness, Authenticity, and Security Challenges", *International Conference on Computer Vision, Graphics, and Image Processing (CVGIP),* Taiwan, Aug., 2025.
4. Invited talk, "Accelerating FHE-based DNN Secure Inference with GPU", Nvidia GTC, San Jose, CA, USA, Mar., 2025.
5. **Keynote**, "Enhanced Learning with Instance-Dependent and Long-Tail Noisy Label Problems", Artificial Intelligence with Biased or Scarce Data (AIBSD) Workshop in conjunction with *AAAI Conference on Artificial Intelligence (AAAI)*, Vancouver, Canada, Feb., 2024.
6. Invited talk, "Trustworthy AI in a Smarter World: Exploring Multi-Modal Media Forensics and Cybersecurity", H&J Global Chair - UI-NYCU Joint AI Lab, National Yang Ming Chiao Tung University, Taiwan, 2023.
7. Invited talk, "Toward Secure, Robust, and Trustworthy Federated Learning", H&J Global Chair - UI-NYCU Joint AI Lab, National Yang Ming Chiao Tung University, Taiwan, 2022.
8. Invited talk, "Research Highlights in Image Inpainting, Federated Learning, and Image Forensics", Academia Sinica, Taiwan, 2022.
9. Seminar, "Scene Text Detection/Recognition and the AICup 2021 Challenge", AICup Challenge Seminar Workshops, Taiwan, 2021.
10. Invited talk, "AI Video Analytic Research in the SUNY Albany CVML Lab", Academia Sinica, Taiwan, Dec. 2019.
11. Invited talk, "AI Video Analytics", National Chung Cheng University, Taiwan, Nov. 2019.
12. **Keynote**, "AI Visual Intelligence Technologies for Building a Smarter World", *IEEE International Conference on Universal Village, Special Session on Future Intelligent Manufacturing*, Massachusetts Institute of Technology, Boston, MA, Oct. 2018.
13. S. Lyu and M.-C. Chang, "DeepFake the Menace", *Preparing for the Future of Disinformation Workshop*, MIT Media Lab, Boston, MA, Sep. 2018.
14. Invited talk, "Video Analytics for AI City Smart Transportation", *Advanced Computer Vision Workshop* at CITI, Academia Sinica, Taiwan, 2017.
15. Invited seminar, "Video Analytics for Automatic Situational Awareness", University of Electronic Science and Technology of China (UESTC), Chengdu, China, 2015.
16. Demo Presentation, J. Chen, M.-C. Chang, and P. Tu, "A Live Face Analysis System in Public Waiting Area", *IEEE Automatic Face and Gesture Recognition (FG)* demo session, Ljubljana, Slovenia, 2015.
17. Invited seminar: "Emotion Detection Technologies and Their Application in Classroom and Lab Research", (with Jixu Chen and Reza Feyzi), School of Education, University at Albany, State University of New York, Albany, New York, 2014.

18. Invited seminar: “Unconstrained Visual Social Interaction Analysis in Groups”, National Taiwan University, Taipei, Taiwan, 2014.
19. Invited seminar, “Advanced Video Surveillance and Group Scenario Recognition”, National Tsing Hua University, Hsinchu City, Taiwan, 2012.
20. Seminar, “From Computer Vision to Visual Intelligence”, Research Center for Information Technology Innovation, Academia Sinica, Taipei, Taiwan, 2010.
21. Poster Presentation & Demo, “Regularizing 3D Medial Axis Using Medial Scaffold Transforms”, IEEE Computer Vision and Pattern Recognition (CVPR), Anchorage, Alaska, USA, 2008.
22. Seminar, GE Global Research, Niskayuna, NY, USA, 2008.
23. Oral Presentation & Demo, “Surface Reconstruction from Point Clouds by Transforming the Medial Scaffold”, IEEE 3-D Digital Imaging and Modeling (3DIM), Montreal, Canada, 2007.
24. Seminar, “Surface Reconstruction and Medial Axis Computation from Unorganized Points”, Computer Vision Reading Group, LEMS, Engineering, Brown University, Providence, RI, USA, 2007.
25. Oral Presentation, “3D Shape Registration using Regularized Medial Scaffolds”, IEEE 3D Data Processing Visualization and Transmission (3DPVT), Thessaloniki, Greece, 2004.
26. Seminar on Recent Research on Medial Axis Based Shape Representations and Applications, Department of Computer Science and Information Engineering, National Taiwan University, Taipei, Taiwan, 2004.
27. Oral Presentation, “IC Print Mark Quality Inspection Algorithms”, IPPR Conference on Computer Vision, Graphics and Image Processing (CVGIP), Taichung, Taiwan, 1997.

Professional Service

Professional Appointments

1. **Reviewer, International Conference on Learning Representations (ICLR) 2027 Workshop Proposals.**
2. **Associated Editor**, IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 10/2025 to 12/2027.
3. **Area Chair**, Neural Information Processing Systems (NeurIPS) Evaluations and Datasets Track, 2026.
4. **Area Chair**, International Conference on Learning Representations (ICLR), 2026.
5. **Area Chair**, Computer Vision and Pattern Recognition (CVPR) Conference, 2026.
6. **Senior PC,** Association for the Advancement of Artificial Intelligence (AAAI) Conference, 2025-2026.
7. **Technical Committee**, IEEE Consumer Technology Society (CTSoc), 3/2025.
8. **Associated Editor**, IEEE Transactions on Multimedia, 10/1/2024 to 9/30/2026.
9. **TPC Co-Chair**, IEEE International Conference on Multimedia Expo (ICME), 2025-2026.
10. **General Co-Chair,** IEEE Conference on Advanced Video and Signal-based Surveillance (AVSS), 2025.
11. **General Co-Chair Lead**, IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR), 2024.
12. **Program Co-Chair Lead**, IEEE Conf. on Advanced Video and Signal-based Surveillance (AVSS), 2024.
13. **Chair, Steering Committee,** IEEE Conf. on Advanced Video and Signal-based Surveillance (AVSS), 2022-present.
14. **TPC Chair Lead**, IEEE Conference on Multimedia Information Processing and Retrieval (MIPR), 2022.
15. **Program Chair**, IEEE Conference on Advanced Video and Signal-based Surveillance (AVSS), 2019.
16. **Area Chair**, IEEE International Conference on Multimedia & Expo. (ICME), 2021-2024, 2026. I was selected as one of the outstanding area chairs of ICME 2021.
17. **Area Chair**, IEEE International Conference on Image Processing (ICIP), 2017, 2019-2026.
18. **Co-Chair**, AI City Challenge Workshop in conjunction of the IEEE Computer Vision and Pattern Recognition (CVPR), 2018-2022 and on-going.
19. **Organization Committee**, Low-Power Computer Vision (LPCV) Workshop and Challenge in conjunction of the CVPR 2020 Workshop, ICCV 2021 Workshop, and on-going.
20. **Co-Chair**, International Workshop on the Traffic and Street Surveillance for Safety and Security (IWT4S), in conjunction with AVSS 2017-2019.
21. **Co-Chair**, Application Track, NVIDIA AI City Challenge in conjunction of the IEEE Smart World Congress, San Jose, CA, USA, 2017.

22. **Publication and Associate Chair**, IEEE Conference on Advanced Video and Signal-based Surveillance (AVSS), Lecce, Italy, 2017.
23. **Steering Committee,** IEEE Conference on Advanced Video and Signal-based Surveillance (AVSS), 2017-present.

Journal Referee:

1. ACM Trans. on Multimedia Computing, Communications, and Applications (TOMM): 2022-2023.
2. ACM Trans. On Multimedia (TMM): 2023.
3. ASME Journal of Computing and Information Science in Engineering (JCISE): 2010.
4. Computers, Materials & Continua (CMC): 2023.
5. Elsevier Big Data Research (BDR): 2020.
6. Elsevier Computers and Electronics in Agriculture: 2022.
7. Elsevier Computers & Electrical Engineering: 2022.
8. Elsevier Computers & Mathematics with Applications: 2009.
9. Elsevier Computer Vision and Image Understanding: 2023.
10. Elsevier Digital Signal Processing: 2023.
11. Elsevier Engineering Applications of Artificial Intelligence (EAAI): 2023.
12. Elsevier Expert Systems with Applications (ESWA): 2022-2023.
13. Elsevier Image and Vision Computing (IVC): 2009.
14. Elsevier Journal of Sustainable Cities and Society: 2022.
15. Elsevier Knowledge-Based Systems: 2023.
16. Elsevier Measurement: 2022-2023.
17. Elsevier Pattern Recognition: 2023.
18. Elsevier Signal Processing: Image Communication: 2020,2023.
19. Elsevier Sustainable Cities and Society: 2022.
20. Elsevier Transportation Research Part D: Transport and Environment: 2022.
21. Elsevier Visual Informatics: 2022.
22. IEEE Trans. on Circuits and Systems for Video Technology (TCSVT): 2021-2022.
23. IEEE Trans. on Cybernetics: 2021.
24. IEEE Trans. on Emerging Topics in Computational Intelligence (TETCI), 2022.
25. IEEE Trans. on Image Processing (TIP): 2009-2010, 2023-2024.
26. IEEE Trans. on Intelligent Transportation Systems (ITS): 2020.
27. IEEE Trans. on Multimedia, 2011, 2020, 2022, 2024.
28. IEEE Trans. on Neural Networks and Learning Systems (TNNLS): 2022.
29. IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI): 2022.
30. IET Computer Vision: 2013.
31. Nature Scientific Reports: 2023.
32. Journal of Applied Analysis and Computation (JAAC), 2015.
33. Springer International Journal of Computer Vision (IJCV): 2009, 2016.
34. Springer Multimedia Tools and Applications (MTA): 2022-2023.
35. Springer Machine Vision Applications (MVA): 2010-2011.
36. Springer The Visual Computer Journal (TVCJ): 2009, 2011-2013, 2021-2022.
37. Springer Numerical Algorithms (NUMA): 2022.
38. Springer Multimedia Tools and Applications (MTAP): 2021-2022.

Conference Technical Committee and Referee:

1. International Conference on Learning Representations (ICLR) Workshop Proposal Reviewer: 2027
2. IEEE Conference on Computer Vision and Pattern Recognition (CVPR): 2011, 2013, 2016, 2019-2024.
3. IEEE International Conference on Computer Vision (ICCV), 2013, 2015, 2019, 2021, 2023.
4. European Conference on Computer Vision (ECCV): 2010, 2012, 2014, 2016, 2020, 2024.
5. Conference on Neural Information Processing Systems (NeurIPS), 2021-2024.
6. International Conference on Machine Learning (ICML), 2022-2024.
7. International Conference on Learning Representations (ICLR), 2022-2025.

8. Association for the Advancement of Artificial Intelligence (AAAI), 2020-2025.
9. International Joint Conference on Artificial Intelligence (IJCAI): 2019-2024.
10. The British Machine Vision Conference (BMVC), 2021.
11. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV): 2023-2025.
12. Int'l Conf. on Medical Image Computing & Computer Assisted Intervention (MICCAI), 2021-2024.
13. Asian Conference on Computer Vision (ACCV): 2022.
14. International Conference on Pattern Recognition (ICPR): 2018, 2022.
15. IEEE Conf. on Adv. Video and Signal-based Surveillance (AVSS), 2010-2011, 2014, 2018, 2019-2021.
16. IEEE International Conference on Robotics and Automation (ICRA), 2022.
17. International Conference on Machine Learning and Data Mining (MLDM), 2022.
18. International Conference on Computer Communication and the Internet (ICCCI), 2023.
19. International Conference on Image Processing and Vision Engineering (IMPROVE), 2021-2022.
20. International Conference on Computer Vision Theory and Applications (VISAPP), 2021.
21. International Workshop on Smart Sensor Networks, 2012.
22. International Conference on Biomedical Engineering and Biotechnology (iCBEB), 2013-16.
23. International Conference on Electronics, Communications and Network (CECNet), 2014-2015.
24. IEEE Int'l Conference on Computer Graphics, Vision and Information Security (CGVIS), 2015.
25. International Conference on Image Information Processing (ICIIP): 2013.

University, College, and Departmental Service Activities

- Search Committee, UAlbany Faculty Cluster Hire in Artificial Intelligence, 2023.
- Artificial Intelligence and Cybersecurity institute design team, 2021.
  Role: Invited and appointed by the President as an expert to develop a schema for structure and organization of the institute, and its correction to the university as well as startup needs and long-term funding model at large.
- UAlbany Honor College scholarship committee, 2017-2020.
  Role: review student applications and guide student development with constructive comments.
- CS graduate admission committee, 2017-2020, and chair of committee 2021.
- CS undergraduate curriculum committee, 2019.
- ECE curriculum committee, 2017.

Professional Organization Memberships

Senior Member, IEEE	2019-present
Full Member, Sigma Xi, the Scientific Research Society	2009-2011
Member, IEEE and IEEE Computer Society	2008-2011, 2016-
Student Member, Society for Industrial and Applied Mathematics (SIAM)	2003-2008
President, Brown Taiwanese Student Association	2002-2003
Member, Chinese Image Processing and Pattern Recognition Society (CIPPR), International Association for Pattern Recognition (IAPR) society.	1997-1998

Media Coverage

1. PBS NewsHour Interview on AI image generators, deep learning models and their impacts, 2022.
2. GE Research Funds Rail Yard Project – Yard Locomotive Detection System Development, **SUNY Albany News**, 2020.
3. 'Deep fake' videos that can make anyone say anything worry U.S. intelligence agencies, **Fox News**, 2019.
4. Exposing Fake Videos, **SUNY Albany News**, 2018.
5. Deep into Deep Learning, **SUNY Albany News**, CEAS Faculty and Students Win Award in First IEEE AI City Challenge, **SUNY Albany News**, 2017.
6. *Breakthrough – More than Human: the Stories behind the Science – GE*. This six-part TV documentary

series developed by GE and **National Geographic Channel** directed by Paul Giamatti focusing on scientific progress and innovation is aired Sunday 9pm Nov. 8, 2015, on the NatGeo Channel.

7. Engines of Inference: Peter Tu's Tech Uses Computer Vision to Understand Emotions, Human Behavior, GE Reports, 2015.
8. *GE Scientists Develop Multi-Sensing Handheld Probe to Assess and Prevent Pressure Ulcers from Forming During Hospital Stays*,
   Yahoo Finance News, March 19, 2015, **BusinessWire.**
9. *Public-Private Partnership Aims to Improve Use of Public Surveillance Cameras in Law Enforcement*, **SUNY Albany News**, Dec. 2014.
10. *Computers That See You, Read You and Even Tell You to Wash*, **New York Times**, Jan. 2, 2011.

Teaching

- **Artificial Intelligence** CSI-435/535: Fall 2017, Fall 2018, Fall 2019, Fall 2020, Fall 2021, Fall 2022, Fall 2025, Spring 2026.
- **Computer Vision** CSI-471/571: Spring 2025.
- **Machine Learning** CSI-536: Spring 2022, Spring 2023.
- **Data Mining** CSI-431/531: Spring 2020, Fall 2022, Spring 2023, Spring 2025.
- **Algorithms and Data Structures** CSI-503: Spring 2025, Spring 2026.
- **Computer Graphics** CSI-502: Spring 2019.
- **Image Processing** CEN-360: Spring 2018.
- **Human Computer Interaction** CEN-470: Spring 2017.
- **Microprocessor Applications** CEN-353: Fall 2016.
- **Special Topics in AI and Cybersecurity** National Taiwan University: Fall 2022, Fall 2023.
- **Trustworthy Artificial Intelligence** National Taiwan University: Fall 2024.
- **Frontier R&Ds in Image/Video Processing and Industrial Computer Vision** University of Chinese Academy of Sciences (UCAS), Summer 2015.
- **Artificial Intelligence II – Computer Vision** CSI-635: Spring 2013.
  Design and teach a new course at SUNY Albany.

Honors and Certificate

2010, 2018	Human Research Curriculum Completion, Collaborative Institutional Training Initiative.
12/2009	Six Sigma Green-Belt Certificate, General Electric Company.

Professional Skills

- 25+ years of programming experience in C/C++ (VXL, OpenCV, PCL, Vtk/Itk, VLFeat, Caffe, OpenInventor, OpenGL, MFC/.NET/wxWindows/Qt, Boost); Python, Matlab/Maple; Java, Pascal, Fortran, Basic; Assembly, SIMD (SSE); Script languages (Unix shell), Web (Html), etc., on PC, Unix, Mac.
- Expert in object-oriented programming (OOP), rapid application development, C++ class libraries, Windows programming, design pattern, multi-programming, threading.
- Contributed to the development of high-quality C++ code in VXL / LEMSVXL computer vision library (http://vxl.sourceforge.net, http://www.lems.brown.edu/vision/software/vxl), including the UAlbany VXL library (uavxl), the Brown mesh library (brl/bmsh), 2D/3D medial axis and shock computation libraries, other geometry computation, registration (ICP), matching, visualization, animation, and GUI libraries.

Other Activities

11/2014-present	Board Member, Treasurer (2016-), Taiwanese American Association (TAA) of Capital District of New York.
9/1994-9/1995	Leader, Service Section of the Fundamental Culture Service Club, National Taiwan University.

7/1994-8/1994 Captain, Summer College Students' Social Service Team, National Taiwan University.